Unified Biquaternion Theory (UBT)
Učebnice a průvodce pro inženýry

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21. srpna 2026

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Obsah

Předmluva: jak tuto učebnici číst

Tato kniha je pracovní učebnicový a referenční text programu Unified Biquaternion Theory (UBT). Jejím prvním úkolem je podat matematiku natolik srozumitelně a souvisle, aby bylo možné později znovu rekonstruovat vazby mezi jednotlivými větvemi UBT bez spoléhání na paměť nebo staré chaty. Proto jsou v textu záměrně odvozovány i klasické matematické výsledky, pokud jsou potřebné pro velký obraz UBT, přestože samotná matematika může být desítky či stovky let známá.

Učebnice rozlišuje tři vrstvy:

1. **klasické základy**: standardní matematika a fyzika s dostatečným odvozením, aby byl text co nejvíce soběstačný;
2. **UBT mosty**: kontrolované převody těchto standardních struktur do UBT notace, omezených modelů nebo konkrétních hypotéz;
3. **UBT tvrzení a otevřené gapy**: výroky závislé na kanonickém poli UBT, jeho akci, dynamice nebo empirickém programu.

Klasická identita se nestává novým výsledkem jen proto, že ji UBT používá. Stejně tak ale skutečnost, že jde o známou matematiku, není důvod ji z učebnice vyhodit: výzkumná hodnota může spočívat v tom, jak jsou známé struktury složeny dohromady a jakou novou otázku pro UBT jejich spojení přesně formuluje. Kanonické claim dokumenty proto zůstávají konzervativní, zatímco učebnice může být široká a didaktická, pokud jsou statusy tvrzení vždy zřetelné.

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Primární studentská edice je česká. Anglická verze může být později vedena jako paralelní edice se stejnou matematickou strukturou a stejnými statusy tvrzení; nové kapitoly se nemají psát jako náhodná směs češtiny a angličtiny.

Každý živý zdrojový soubor zároveň používá strojově čitelnou AI-provenance politiku repozitáře. Tato učebnice je pracovní materiál Tier C, dokud lidský editor výslovně nezmění tier v autoritativní provenance mapě; AI agent nesmí provádět vlastní povýšení nebo atestaci.

Odvození v paperech a teorémově důležité části učebnice se mají nezávisle kontrolovat podle `docs/DERIVATION_VERIFICATION_POLICY.md`. Pro formalizovatelnou přesnou matematiku je preferovaný Lean; vedle něj se používají nezávislé CAS a numerické kontroly. Úspěšný výpočet v nástroji však ověřuje jen to, co bylo do nástroje skutečně zakódováno, a sám o sobě nemění fyzikální status UBT tvrzení.

Část I

Základy a geometrie

Kapitola 1

UBT za hodinu: motivace, architektura a současný stav

1.1 K čemu tato kapitola slouží

Unified Biquaternion Theory (UBT) asks whether spacetime geometry and internal field structure can be encoded in one complex-quaternion-valued master field, rather than introduced as unrelated fundamental objects. The active starting notation is

$$\Theta(q,\tau) \in \mathbb{B} = \mathbb{C} \otimes \mathbb{H}, \qquad \tau = t + i\psi.$$

The notation is compact, but it hides several logically different questions: what is defined, what is kinematically representable, what follows from an action, and what is only a research hypothesis. The textbook keeps those levels separate throughout.

1.2 Současný lokální geometrický řetězec

The clean local GR branch starts from the covariant derivative of the single field,

$$E_\mu := \mathcal{N}_0^{-1/2} D_\mu \Theta.$$

On the real Lorentz slice, write

$$E_\mu = i e_\mu{}^0 \mathbf{1} + e_\mu{}^k \mathbf{e}_k, \qquad e_\mu{}^a \in \mathbb{R}.$$

Quaternion conjugation is denoted by $\sharp$. The symmetrised product is central and defines the metric,

$$\boxed{\frac{1}{2}\left(E_\mu^\sharp E_\nu + E_\nu^\sharp E_\mu\right) = g_{\mu\nu}\mathbf{1}.}$$

Thus the minimal local metric construction does not require an observer projection, a compact-fibre average, or an embedding map. This replaces an older architectural picture preserved in the historical archive.

The same first derivative behaves as a tetrad. For a nondegenerate tetrad the map $e_\mu{}^a \mapsto g_{\mu\nu}$ has the familiar counting

$$16 - 6 = 10,$$

because six local Lorentz changes leave the ten metric components invariant. This resolves the local rank problem at the kinematic level.

1.3 Konexe, křivost a co je stále dynamické

A local Lorentz/biquaternionic connection enters D_μ. In a torsion-free GR branch the compatible spin connection is the Levi–Civita spin connection computed from the tetrad. With torsion,

$$\omega = \mathring{\omega}(e) + K(T),$$

so specifying the tetrad and torsion determines the metric-compatible connection. Curvature belongs to the commutator of covariant derivatives, not to the algebraic commutator of the tetrad legs:

$$[D_\mu, D_\nu] \quad \text{contains curvature.}$$

The present GR work closes important local representability questions, but it does not yet derive all Einstein dynamics from the original UBT master action. In particular, action selection, physical torsion constraints, the full perturbation bridge, and global continuation remain status-controlled open problems. The authoritative wording is maintained in `STATUS.md` and `WHAT_IS_PROVED.md`.

1.4 Komplexní čas a theta/spektrální větev

The canonical notation retains

$$\tau = t + i\psi,$$

with ψ interpreted in the active complex-time branch as an internal phase coordinate rather than a second macroscopic clock. Reduced theta models then naturally contain factors of the form

$$S(t, \psi) = \sum_n a_n e^{-\pi\psi n^2} e^{i\pi t n^2}.$$

This one expression admits several standard representations: Fourier in t, Laplace or Mellin in ψ, Jacobi/Poisson duality, and, in controlled quadratic examples, a relation between mode sums and path/winding sums. Those classical tools are developed explicitly later because they organise several otherwise disconnected UBT research tracks.

1.5 Kalibrační a částicové sektory

The algebra $\mathbb{C} \otimes \mathbb{H} \cong M_2(\mathbb{C})$ naturally supports left and right actions, spinorial Lorentz transformations, and internal symmetry constructions. Some gauge-sector statements are proved or conditional at stated levels; others, including parts of hypercharge, generation selection, and dynamical symmetry breaking, remain open. This textbook therefore never infers a full Standard Model derivation merely from the existence of the biquaternion algebra.

1.6 Konstanta jemné struktury: současné bezpečné tvrzení

The alpha programme contains a reproducible structural mechanism selecting the integer 137 under stated assumptions, including a prime-stability result. However the overall track is currently not a first-principles derivation of the physical fine-structure constant. Two blockers are especially important:

1. the effective coefficient required by the winding potential has not been derived uniquely from the UBT action (Gap G137-B);
2. the audited loop multiplicity and the twist-sector multiplicity are not yet identified by a proved bridge.

Consequently neither the phrase "fit-free alpha derivation" nor the numerical value 137.036... should be presented here as a closed UBT prediction. Chapter 7 derives the conditional mathematics and states the gaps at every step.

1.7 Jak číst statusové značky

The repository uses two independent bookkeeping systems that must not be confused.

First, scientific claim levels distinguish standard identities, proved UBT lemmas, conditional results, motivated conjectures, observations, and open gaps. Second, AI provenance tiers record editorial/review status of files. The latter are defined by `PROVENANCE_TIERS.yaml`: Tier A is author-attested, Tier B machine-verified, Tier C working, and Tier D historical. This textbook is Tier C working material. Its role is to explain and connect the mathematics, not to silently promote a claim to Tier A.

1.8 Pravidlo autoritativního zdroje pro učebnici

The source map in `docs/textbook/SOURCE_MAP.md` records where each chapter gets its live status. Historical material in `ARCHIVE/` can be valuable for understanding how an idea evolved, but it is never silently `\input` as the current theory.

Kapitola 2

Matematický základ: bikvaterniony bez slz

This chapter introduces the algebra needed by later chapters. Detailed Lorentz geometry is derived self-contained in Chapter 3; the live textbook does not import the historical consolidation archive.

Kapitola 3

Lorentzova geometrie na hermitovském řezu $\mathbb{H}_{\mathbb{C}}$

3.1 Klasický základ: $\mathrm{Herm}(2,\mathbb{C})$

This section is standard spinor/Lorentz mathematics. It is included because UBT uses the same algebra through the isomorphism

$$\mathbb{B} = \mathbb{C} \otimes \mathbb{H} \cong M_2(\mathbb{C}).$$

Let σ_i be the Pauli matrices and associate a real four-vector $x = (x^0, x^1, x^2, x^3)$ with the Hermitian matrix

$$X = x^0 I_2 + x^i \sigma_i = \begin{pmatrix} x^0 + x^3 & x^1 - ix^2 \\ x^1 + ix^2 & x^0 - x^3 \end{pmatrix}.$$

A direct determinant calculation gives

$$\boxed{\det X = (x^0)^2 - (x^1)^2 - (x^2)^2 - (x^3)^2.}$$

Thus the determinant on the Hermitian slice is exactly the Minkowski quadratic form with signature $(+ - --)$.

3.2 Akce $SL(2,\mathbb{C})$

For $A \in SL(2,\mathbb{C})$ define

$$X \longmapsto X' = AXA^\dagger.$$

Hermiticity is preserved because

$$(AXA^\dagger)^\dagger = AXA^\dagger.$$

The determinant is also preserved:

$$\det X' = \det A \, \det X \, \det A^\dagger = \det X,$$

since $\det A = 1$ and $\det A^\dagger = \overline{\det A} = 1$. Therefore the induced real linear transformation of the four coefficients preserves the Minkowski norm. Restricting continuously to the identity gives a homomorphism

$$SL(2,\mathbb{C}) \longrightarrow SO^+(1,3).$$

Both A and $-A$ act identically on X, so the kernel contains $\{\pm I_2\}$. The standard result is the double cover

$$\boxed{SL(2,\mathbb{C})/\{\pm I_2\} \cong SO^+(1,3).}$$

No split-quaternion algebra is required for this construction.

3.3 Převod do bikvaternionové notace

Use the canonical matrix representation

$$\mathbf{e}_i \longmapsto i\sigma_i.$$

Then $\sigma_i \leftrightarrow -i\mathbf{e}_i$, and the Hermitian matrix above corresponds to the biquaternion

$$X_\mathbb{B} = x^0\mathbf{1} - ix^i\mathbf{e}_i.$$

This is simply another coordinate description of the same $\mathrm{Herm}(2,\mathbb{C})$ slice.

UBT's current tetrad convention often uses instead the Lorentz-real basis

$$\mathbf{u}_0 = i\mathbf{1}, \qquad \mathbf{u}_k = \mathbf{e}_k,$$

so a tetrad leg is

$$E_\mu = ie_\mu{}^0\mathbf{1} + e_\mu{}^k\mathbf{e}_k.$$

With quaternion conjugation $\sharp$,

$$E_\mu^\sharp = ie_\mu{}^0\mathbf{1} - e_\mu{}^k\mathbf{e}_k.$$

Using $\mathbf{e}_j\mathbf{e}_k + \mathbf{e}_k\mathbf{e}_j = -2\delta_{jk}$, one obtains

$$\begin{aligned}\frac{1}{2}(E_\mu^\sharp E_\nu + E_\nu^\sharp E_\mu) &= \left(-e_\mu{}^0 e_\nu{}^0 + e_\mu{}^k e_\nu{}^k\right)\mathbf{1}\\ &= g_{\mu\nu}\mathbf{1}.\end{aligned}$$

This is the same Lorentz geometry with the opposite overall signature convention $(-+++)$. A signature convention is bookkeeping; it is not a new physical degree of freedom.

3.4 Proč je to důležité pro UBT

The classical result establishes three useful facts before any UBT dynamics is introduced:

1. complexified quaternions already carry the spinorial Lorentz action;
2. a four-dimensional Lorentzian quadratic form can be read internally from the algebra;
3. left/right multiplication matters, because the Lorentz action is naturally two-sided in matrix language.

What this classical construction does *not* prove is equally important. It does not prove that the UBT master field dynamically generates every desired tetrad, it does not derive Einstein's equations, and it does not select the physical connection. Those are UBT-specific bridge problems addressed in the covariant-tetrad chapter and the canonical GR closure ledger.

3.5 Konkrétní boost jako kontrolní příklad

Take a boost of rapidity χ along the third spatial axis,

$$A = \exp\left(\frac{\chi}{2}\sigma_3\right) = \begin{pmatrix} e^{\chi/2} & 0 \\ 0 & e^{-\chi/2}\end{pmatrix}.$$

Applying $X' = AXA^\dagger$ and comparing coefficients gives

$$\begin{pmatrix} x'^0 \\ x'^3 \end{pmatrix} = \begin{pmatrix} \cosh\chi & \sinh\chi \\ \sinh\chi & \cosh\chi \end{pmatrix}\begin{pmatrix} x^0 \\ x^3 \end{pmatrix}, \qquad x'^1 = x^1, \quad x'^2 = x^2.$$

Then

$$(x'^0)^2 - (x'^3)^2 = (x^0)^2 - (x^3)^2,$$

which is the ordinary Lorentz boost written as an internal algebra action.

3.6 Živé zdroje a historická poznámka

The algebra convention is kept in `canonical/algebra/biquaternion_algebra.tex`; current GR use of the Lorentz-real tetrad is documented in the active GR track. The former `appendix_P6_lorentz_in_HC.te` survives in the historical archive, but the live textbook no longer imports it.

Kapitola 4

Kovariantní derivace, konexe a čistá geometrie UBT

This chapter explains the geometric core of UBT in the form closest to the original intuition of the theory:

$$\Theta \quad \longrightarrow \quad E_\mu := D_\mu \Theta \quad \longrightarrow \quad \{E_\mu, E_\nu\} \text{ and } [D_\mu, D_\nu].$$

The anticommutator carries the metric, while the commutator of covariant derivatives carries curvature. No embedding map, compact-fiber average, or observer projection is needed in the minimal local construction.

4.1 Obyčejná, parciální a kovariantní derivace

For an ordinary scalar function $f(x)$, the derivative df/dx measures how the number f changes when x changes. For a field on spacetime we use partial derivatives,

$$\partial_\mu \Theta := \frac{\partial \Theta}{\partial x^\mu}, \qquad \mu = 0, 1, 2, 3.$$

A partial derivative compares components written in a fixed basis. This is sufficient if the same basis can be used everywhere. Geometry becomes subtler when the local frame itself rotates or boosts from point to point.

A simple analogy is a vector written in a rotating coordinate frame. Even a physically constant vector acquires changing components because the basis is changing. The covariant derivative corrects for this basis change:

$$D_\mu \Theta = \partial_\mu \Theta + \rho_*(\Omega_\mu)\Theta.$$

Here Ω_μ is the connection and ρ_* tells us how the connection acts in the representation of Θ. For biquaternions, a left action, a right action, and a two-sided action are genuinely different and must not be silently interchanged.

4.2 Co je konexe?

A connection is not a force by itself. It is a rule for comparing local frames at neighboring points. The symbol Ω_μ is the UBT internal or biquaternionic-frame connection.

Christoffel symbols,

$$\Gamma^\rho{}_{\mu\nu},$$

are closely related, but they act on coordinate indices. For example,

$$\nabla_\mu V^\rho = \partial_\mu V^\rho + \Gamma^\rho{}_{\mu\sigma} V^\sigma.$$

Thus:

- $\Gamma^\rho{}_{\mu\nu}$ tells us how the coordinate basis changes;
- $\omega_\mu{}^a{}_b$, and its spin/biquaternionic lift Ω_μ, tell us how the local Lorentz frame changes.

They are related by the tetrad compatibility identity

$$\partial_\mu e_\nu{}^a - \Gamma^\rho{}_{\mu\nu} e_\rho{}^a + \omega_\mu{}^a{}_b e_\nu{}^b = 0.$$

They are therefore two descriptions of the same geometric transport, not the same coefficients. In particular, $\Gamma = \mathrm{Re}\,\Omega$ is not a valid identification.

4.3 Proč klasická konexe není další volné pole

For a nondegenerate tetrad, metric compatibility and zero torsion determine a unique Lorentz connection:

$$\boxed{\mathring{\omega}_\mu{}^a{}_b = e_b{}^\nu \left(\mathring{\Gamma}^\rho{}_{\mu\nu}(g) e_\rho{}^a - \partial_\mu e_\nu{}^a \right).}$$

The circle marks the torsion-free Levi–Civita connection. Thus, in the ordinary torsion-free GR branch, the frame connection is computed from the tetrad just as the Christoffel symbols are computed from the metric. A local Lorentz rotation changes the numerical coefficients of the tetrad and the connection but does not change the geometry.

4.3.1 Torze a kontorze

The torsion two-form is

$$T^a = de^a + \omega^a{}_b \wedge e^b.$$

A general metric-compatible connection can be written

$$\boxed{\omega = \mathring{\omega}(e) + K(T),}$$

where K is the contorsion. With $T^a = \frac{1}{2} T^a{}_{bc} e^b \wedge e^c$ and the convention above,

$$\boxed{K_{abc} = \frac{1}{2} \left(T_{cab} - T_{abc} - T_{bca} \right).}$$

This formula has an important conceptual consequence: once the tetrad and the torsion are specified, the metric-compatible connection is unique. The open full-UBT question is therefore not an arbitrary leftover Ω_μ; it is the dynamical law selecting T.

If the UBT vacuum equations imply $T = 0$, the connection reduces to the Levi–Civita spin connection. If spin or another biquaternionic source creates torsion, the same theorem reconstructs the connection from that torsion.

4.4 Plochý prostor a explicitní UBT reprezentant

In flat Minkowski spacetime, using inertial Cartesian coordinates and a constant local frame, one can choose

$$\Gamma^\rho{}_{\mu\nu} = 0, \qquad \Omega_\mu = 0, \qquad D_\mu = \partial_\mu.$$

This statement can be strengthened from a choice of gauge to an explicit single-field representer. Let

$$\boxed{\Theta_{\mathrm{SR}}(x) = \Theta_0 + \sqrt{\mathcal{N}_0} \left(i x^0 \mathbf{1} + x^k \mathbf{e}_k \right).}$$

Then

$$\mathcal{N}_0^{-1/2} \partial_0 \Theta_{\mathrm{SR}} = i\mathbf{1}, \qquad \mathcal{N}_0^{-1/2} \partial_k \Theta_{\mathrm{SR}} = \mathbf{e}_k.$$

The central anticommutator of these four constant directions is the Minkowski metric. Every second spacetime derivative of Θ_{SR} vanishes, so it also solves the standard source-free constant-coefficient second-order master operator in this flat branch.

More generally, for any constant Lorentz tetrad $\bar{E}_\mu$,

$$\Theta_{\mathrm{aff}}(x) = \Theta_0 + \sqrt{\mathcal{N}_0}\,\bar{E}_\mu x^\mu$$

generates that tetrad exactly. This closes the special-relativistic representer problem, although it does not prove uniqueness or curved-space existence.

In polar coordinates, or in an accelerating frame, Γ and Ω may be nonzero even though spacetime is flat. The invariant statement of flatness is vanishing curvature, not vanishing connection coefficients in every frame.

4.5 Čtyři kovariantní gradienty jako tetráda

Define

$$E_\mu := \frac{1}{\sqrt{\mathcal{N}_0}} D_\mu \Theta.$$

The four objects E_0, E_1, E_2, E_3 are the local UBT rulers and clock direction. They are not four extra matter fields; they are the first covariant derivatives of the single fundamental field Θ.

Let i be the ordinary commuting complex unit and $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ the quaternion units. In the classical Lorentz sector write

$$E_\mu = i e_\mu{}^0 \mathbf{1} + e_\mu{}^k \mathbf{e}_k, \qquad e_\mu{}^a \in \mathbb{R}.$$

Quaternion conjugation changes the signs of the three quaternion units but leaves the complex scalar coefficient untouched:

$$E^\sharp_\mu = i e_\mu{}^0 \mathbf{1} - e_\mu{}^k \mathbf{e}_k.$$

4.6 Proč je antikomutátor metrikou

Quaternion multiplication contains a scalar product and a cross product. In the symmetrised product the cross products cancel:

$$\boxed{\frac{1}{2}\left(E^\sharp_\mu E_\nu + E^\sharp_\nu E_\mu\right) = g_{\mu\nu}\mathbf{1}.}$$

The right-hand side contains $\mathbf{1}$, the unit biquaternion. The entire left-hand side is already central: an ordinary real number multiplied by the algebra unit. No trace, real-part map, phase projection, or fiber average is required.

Expanding the coefficients gives

$$g_{\mu\nu} = -e_\mu{}^0 e_\nu{}^0 + e_\mu{}^1 e_\nu{}^1 + e_\mu{}^2 e_\nu{}^2 + e_\mu{}^3 e_\nu{}^3,$$

which is exactly

$$g_{\mu\nu} = e_\mu{}^a e_\nu{}^b \eta_{ab}, \qquad \eta_{ab} = \mathrm{diag}(-1, 1, 1, 1).$$

The minus sign of time comes from $i^2 = -1$.

4.7 Co dělá druhá polovina součinu?

The antisymmetric algebraic part is

$$\Sigma_{\mu\nu} = \frac{1}{2}\left(E^{\sharp}_{\mu}E_{\nu} - E^{\sharp}_{\nu}E_{\mu}\right).$$

It describes an oriented plane or bivector and is a natural carrier for local rotation, boost, and spin information. It is not itself the curvature.

Three expressions must be kept separate:

$$\begin{aligned} \{E_\mu, E_\nu\}_\sharp &\longrightarrow \text{metric},\\ [E_\mu, E_\nu]_\sharp &\longrightarrow \text{algebraic bivector/spin structure},\\ [D_\mu, D_\nu] &\longrightarrow \text{connection curvature or gauge field strength.} \end{aligned}$$

4.8 Křivost z nekomutujících derivací

For a one-sided connection, the curvature has the familiar form

$$\mathcal{R}_{\mu\nu} = \partial_\mu\Omega_\nu - \partial_\nu\Omega_\mu + [\Omega_\mu, \Omega_\nu].$$

The final commutator is essential because biquaternion multiplication is noncommutative. Geometrically, curvature measures how a frame changes after transport around a small closed loop.

The equation $E_\mu = D_\mu\Theta/\sqrt{\mathcal{N}_0}$ creates an additional integrability condition. This is the point at which the representation of the connection becomes decisive.

4.9 Proč starý problém s hodností lokálně mizí

A symmetric 4×4 metric has ten independent components. Looking only at one value of $\Theta \in \mathbb{B} \simeq \mathbb{R}^8$ suggests the misleading comparison $8 < 10$.

The metric is not built from the value of Θ alone. It is built from four covariant derivatives E_μ. In the Lorentz sector these form a real 4×4 tetrad $e_\mu{}^a$ with sixteen components. Six changes are local rotations and boosts that leave the metric unchanged. Therefore

$$16 - 6 = 10.$$

More rigorously, every small symmetric metric variation $h_{\mu\nu}$ is produced by

$$\delta e_\mu{}^a = \frac{1}{2}h_{\mu\rho}e^{\rho a}.$$

The tetrad-to-metric map therefore has rank ten at every nondegenerate tetrad. This resolves the local kinematic rank question. It does not yet prove that every curved tetrad is generated by an on-shell UBT field.

4.9.1 Správný výpočet může odpovědět na špatně položenou architektonickou otázku

The former single-section rank obstruction was mathematically correct in an induced-metric framework where Einstein equations were to be recovered only from normal variations of one embedding-like section. It did not constitute a no-go theorem for the covariant tetrad $E_\mu = D_\mu\Theta/\sqrt{\mathcal{N}_0}$. The compact-fiber construction repaired the former formulation by adding profile directions, but the tetrad formulation shows that this repair was not required by the original architecture.

This distinction motivates a general research rule:

> Before adding fields, modes, dimensions, projections, averages, or auxiliary structures to solve an obstruction, first test whether the obstruction is an artefact of the formulation in which it was derived.

The fiber route remains mathematically consistent in its own framework. Its problem is weak selection and representer redundancy, not an algebraic contradiction.

4.10 Problém integrability

Once the classical connection is reconstructed from the tetrad and torsion, the defining equation becomes schematically

$$E_\mu = \mathcal{N}_0^{-1/2} \left(\partial_\mu \Theta + \rho_*(\Omega_\mu[E,T])\Theta\right).$$

The unknown E occurs on both sides. This is an implicit nonlinear first-order PDE or fixed-point system. Eliminating Ω kinematically does not by itself prove existence or uniqueness of a pair (Θ, E).

4.10.1 Jednostranná překážka vedoucí k plochosti

Suppose the connection acts only from the left,

$$D_\mu^L \Theta = \partial_\mu \Theta + A_\mu \Theta,$$

so that

$$[D_\mu^L, D_\nu^L]\Theta = F_{\mu\nu}^L \Theta.$$

In a torsion-free compatible tetrad branch, antisymmetrising the covariant Hessian gives

$$F_{\mu\nu}^L \Theta = 0.$$

If Θ is invertible as a biquaternion, then

$$F_{\mu\nu}^L = 0.$$

Thus the naive one-sided regular representation cannot generically support curved GR while retaining an invertible Θ and torsion-free tetrad compatibility. Noninvertible stabilizer configurations may evade this result, but they cannot be assumed to represent arbitrary geometries.

4.10.2 Přirozená oboustranná bikvaternionová derivace

Biquaternions naturally admit both left and right multiplication. Consider

$$\boxed{D_\mu \Theta = \partial_\mu \Theta + A_\mu \Theta - \Theta B_\mu.}$$

A direct expansion gives

$$\boxed{[D_\mu, D_\nu]\Theta = F_{\mu\nu}^A \Theta - \Theta F_{\mu\nu}^B.}$$

Torsion-free compatibility therefore requires

$$F_{\mu\nu}^A \Theta = \Theta F_{\mu\nu}^B.$$

For invertible Θ,

$$\boxed{F_{\mu\nu}^A = \Theta F_{\mu\nu}^B \Theta^{-1}.}$$

Unlike the one-sided case, neither curvature must vanish. The field Θ intertwines the left and right curvature representations. This is a genuine structural narrowing of GAP-10I: the two-sided/bimodule derivative is the minimal algebra-native route that avoids the generic flatness obstruction.

4.10.3 Redukce bez nového pole, bez-torzní no-go a lokální pravá inverze s torzí

On the Lorentz slice $W_L = \{X \mid X^\ddagger = -X\}$, preservation of the slice and of its quadratic form reduces the pure pair, modulo a common central one-form that cancels, to

$$\boxed{A_\mu = \Omega_\mu, \qquad B_\mu = -\Omega_\mu^\ddagger.}$$

Thus A_μ and B_μ are not independent gravitational fields in this branch. They are the two module actions of the one spin connection.

With zero torsion, the same involution-compatible derivative makes a nondegenerate solution $D_\mu\Theta = \sqrt{\mathcal{N}_0}E_\mu$ imply

$$\boxed{\mathring{\nabla}_\mu V^\nu = \delta_\mu{}^\nu,} \qquad \mathcal{L}_V g = 2g.$$

The non-flat Schwarzschild vacuum exterior with nonzero mass has no such proper homothety. The pure pair is therefore closed kinematically and its $K = 0$ branch is closed as a torsion-free no-go.

The qualifier matters. On a sufficiently small non-null Gaussian patch, choose $\rho \neq 0$ with $g^{\mu\nu}\partial_\mu\rho\partial_\nu\rho = \varepsilon = \pm 1$ and set

$$V^\mu = \varepsilon\rho\nabla^\mu\rho, \qquad W_{\mu\nu} = g_{\mu\nu} - \mathring{\nabla}_\mu V_\nu,$$

$$\boxed{K_{\nu\mu\rho} = \frac{W_{\mu\nu}V_\rho - V_\nu W_{\mu\rho}}{V^2}.}$$

Then $K_{\nu\mu\rho} = -K_{\rho\mu\nu}$ and $\nabla_\mu^{(\mathring{\Gamma}+K)}V^\nu = \delta_\mu{}^\nu$. Therefore

$$\Theta = \sqrt{\mathcal{N}_0}V^a\mathbf{u}_a$$

obeys $D_\mu\Theta = \sqrt{\mathcal{N}_0}E_\mu$. Every smooth tetrad has such a local representer with composite metric-compatible contortion and no independent A_μ, B_μ fields.

A relative term $P_\mu X - XQ_\mu$ is no longer required for local kinematic existence. It remains a possible torsion-free completion and, if used, must be composite or auxiliary rather than a new propagating field. A common central constant does not remove the torsion-free obstruction because it cancels.

4.11 Podmíněná dynamická dílčí uzavření

4.11.1 Algebraická Cartanova rovnice torze

In the minimal first-order Hilbert–Palatini branch, independent variation of the Lorentz connection gives

$$\epsilon_{abcd}T^a \wedge e^b = \kappa\tau_{cd}.$$

For a nondegenerate tetrad this is a 24×24 pointwise linear map of rank 24. Thus zero spin current gives $T = 0$, while a specified spin current gives a unique torsion and contorsion. This closes the torsion algebra inside the minimal branch, not the derivation of that branch from fundamental UBT.

4.11.2 Propagace Lorentzova řezu jako množiny pevných bodů

The anti-linear involution

$$\mathcal{J}X = -\overline{X^\sharp}$$

has the physical Lorentz module as its fixed set. If a well-posed unique Cauchy evolution is $\mathcal{J}$-equivariant and its data and sources are fixed by $\mathcal{J}$, uniqueness implies that the solution remains in the Lorentz slice. The final UBT action must still be checked against these hypotheses.

4.11.3 Stabilita metriky v imaginárním čase

A vertical flow

$$\partial_\psi e_\mu{}^a = \Lambda_\psi{}^a{}_b e_\mu{}^b, \qquad \Lambda_{\psi ab} = -\Lambda_{\psi ba},$$

is tangent to a local Lorentz gauge orbit and satisfies $\partial_\psi g_{\mu\nu} = 0$ identically. A second sufficient mechanism is a unique ψ-translation-invariant evolution with ψ-independent initial data. Selecting the physical mechanism and excluding unstable non-gauge modes remains a dynamical task.

4.11.4 Rozšířená holonomie pro předepsané zakřivené koeficienty

For prescribed E_μ, A_μ, B_μ, the inhomogeneous system

$$\partial_\mu \Theta + A_\mu \Theta - \Theta B_\mu = \sqrt{\mathcal{N}_0} E_\mu$$

becomes homogeneous parallel transport for $Y = (\Theta, 1)^T$ in an augmented bundle. A single-valued solution with initial value Y_0 exists exactly when the augmented holonomy fixes Y_0; flat augmented curvature is sufficient on a simply connected domain. This closes prescribed-coefficient integrability, not self-consistent UBT selection of the coefficients.

4.11.5 Podmíněný Einsteinův–Lambda koncový bod

The minimal Palatini branch with zero spin current yields

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}.$$

Under the ordinary four-dimensional Lovelock assumptions, every local natural divergence-free second-order metric equation without extra light geometric fields has gravitational part $aG_{\mu\nu} + bg_{\mu\nu}$. Therefore the conditional low-energy endpoint is unique. UBT must still derive those assumptions, coefficients and matter terms from its canonical action.

4.12 Implicitní versus transcendentní rovnice

The words *implicit* and *transcendental* describe different properties.

An equation is implicit when the unknown occurs inside the map that is supposed to determine it:

$$E = \mathcal{F}(E, \Theta).$$

An equation is transcendental when it contains nonalgebraic functions such as an exponential, logarithm, trigonometric function, or Jacobi theta function:

$$E = e^{-E}.$$

The UBT tetrad equation is already implicit and differential. If the allowed field $\Theta(q, \tau)$ is restricted to a Jacobi-theta or another transcendental function class, then the complete system may be both implicit and transcendental. In a formal paper these properties should be stated separately, even though intuitively both make the unknown "entangled with itself."

4.13 Derivace versus variace

A derivative such as $D_\mu \Theta$ describes how one field configuration changes through spacetime. A variation compares nearby possible configurations:

$$\Theta_\varepsilon = \Theta + \varepsilon\, \delta\Theta.$$

The variation is

$$\delta\Theta = \frac{d\Theta_\varepsilon}{d\varepsilon}\bigg|_0 .$$

Variations are needed when deriving field equations from an action. They are not replacements for spacetime derivatives. When the connection depends on the field,

$$\delta(D_\mu\Theta) = D_\mu(\delta\Theta) + \rho_*(\delta\Omega_\mu)\Theta$$

and the induced connection variation must be included.

4.14 Co je uzavřeno a co zůstává otevřené

The following results are closed at their stated levels:

1. the central anticommutator metric and rank-ten tetrad map;
2. GAP-10Ω-KIN/GR: reconstruction of the compatible connection and its Levi–Civita torsion-free branch;
3. GAP-10T-PALATINI (conditional): the minimal Cartan torsion equation is algebraic and invertible;
4. GAP-10L-CONN and GAP-10L-SYM (conditional): compatible transport and unique equivariant evolution preserve the Lorentz slice;
5. GAP-10I-SR and GAP-10I-PRESCRIBED: the affine flat representer and the exact augmented-holonomy criterion for prescribed curved coefficients;
6. GAP-10I-1S: the naive one-sided invertible curved route is a conditional no-go;
7. GAP-10I-PAIR-KIN: the pure compatible pair reduces to the one spin connection;
8. GAP-10I-PAIR-GR: the $K = 0$ pure pair is a concurrent-vector no-go for the non-flat Schwarzschild vacuum exterior with nonzero mass;
9. GAP-10I-TORSION-LOCAL: every smooth tetrad has a local single-Θ representer with explicit composite metric-compatible contortion;
10. GAP-10D-PALATINI/UNIQUENESS (conditional): the minimal first-order branch and Lovelock hypotheses select Einstein–Λ;
11. GAP-10ψ-KIN/SYM: Lorentz-gauge flow or translation symmetry protects the classical metric under the stated hypotheses.

The undivided full-theory gaps are narrowed rather than fully closed:

1. GAP-10T-SPIN is closed conditionally for the direct fixed-background Palatini variation, while GAP-10T-FLAT-NOGO and GAP-10T-PAIRING-NOGO are closed as no-go results;
2. GAP-10T-DYN: compute the full composite Θ variation and derive the selected first-order branch, a canonical torsion cancellation or translational/relative completion, and normalization from canonical UBT;
3. GAP-10I-CURVED: local kinematics is closed; derive canonical action-level selection and physical admissibility of the composite torsionful branch, or an optional torsion-free relative bimodule branch, and establish regularity and global continuation of (Θ, E, A, B, T);

4. GAP-10L-DYN: prove equivariance and well-posed uniqueness for the final complete dynamics;
5. GAP-10D: derive the low-energy Palatini/Lovelock assumptions, couplings and matter action from the fundamental theory;
6. GAP-10ψ: select the stability mechanism and exclude physical unstable non-gauge modes.

GAP-B-MASTER and GAP-U2Θ remain open. The earlier compact-ψ fiber closure remains a mathematically consistent historical candidate completion, but its weak selection and representer redundancy keep it outside the minimal canonical route.

Část II

Komplexní čas a spektrální struktura

Kapitola 5

Pokračování do komplexního času a Wardovy identity

5.1 Proč tato kapitola zůstává v učebnici

Older alpha work used a “CT scheme” built around the analytic continuation $\tau = t + i\psi$. The archived implementation is not a current source of record, but the underlying mathematics—analytic continuation, gauge identities, and the real-time limit—is standard and useful. This chapter therefore teaches those ingredients and then states exactly what they do and do not establish for UBT.

5.2 Komplexní čas jako parametr

For an analytic quantity $F(\tau)$,

$$\tau = t + i\psi$$

allows one to move between oscillatory and damped descriptions. For an energy eigenmode, for example,

$$e^{-iE\tau/\hbar} = e^{-iEt/\hbar}e^{+E\psi/\hbar},$$

with the sign of the damping determined by the convention for the imaginary direction. In the theta convention used later,

$$e^{-\pi(\psi-it)n^2} = e^{-\pi\psi n^2}e^{i\pi tn^2}.$$

Nothing in this analytic identity by itself creates a new renormalisation constant or fixes a physical coupling.

5.3 Wardova–Takahashiho identita: standardní argument

In QED let $S(p)$ be the full fermion propagator and $\Gamma^\mu(p+q,p)$ the full electromagnetic vertex. Gauge invariance gives the Ward–Takahashi identity

$$q_\mu\Gamma^\mu(p+q,p) = S^{-1}(p+q) - S^{-1}(p).$$

Taking the limit $q \to 0$ yields

$$\Gamma^\mu(p,p) = \frac{\partial S^{-1}(p)}{\partial p_\mu}.$$

At the level of renormalisation constants this enforces

$$\boxed{Z_1 = Z_2,}$$

provided the regulator and subtraction prescription preserve gauge symmetry. This is a standard gauge-theory result. A complex-time continuation may use it only after demonstrating that its contour/regulator preserves the hypotheses required by the identity.

5.4 Transverzalita vlastní energie fotonu

The same gauge symmetry implies

$$q_\mu \Pi^{\mu\nu}(q) = 0,$$

so the vacuum polarisation has the form

$$\Pi^{\mu\nu}(q) = \left(q^2 g^{\mu\nu} - q^\mu q^\nu\right) \Pi(q^2)$$

(up to convention). This is why charge renormalisation can be expressed in terms of the transverse photon two-point function. Again, this is a standard field-theory constraint, not a UBT-specific theorem.

5.5 Kontrolovaný UBT most

A UBT complex-time renormalisation construction would need to establish, for its actual action and contour:

1. analyticity sufficient to continue between the chosen ψ contour and the real-time slice;
2. preservation of the relevant gauge/BRST identities;
3. a regulator and counterterm scheme compatible with those identities;
4. continuity to standard QED in the appropriate real-time limit;
5. an explicit calculation of any finite CT-specific remainder rather than setting it by definition.

Only after these steps could a CT-specific factor be promoted from a modelling choice to a derived quantity.

5.6 Proč se archivní text $\mathcal{R}_{\mathrm{UBT}} = 1$ nepřebírá

The historical alpha branch stated a two-loop baseline $\mathcal{R}_{\mathrm{UBT}} = 1$ under assumptions labelled A1–A3 and then called the resulting alpha baseline "fit-free". Current Tier-A status documents no longer allow that wording to stand as a completed alpha derivation: the coefficient bridge G137-B is open and the effective multiplicity has an audit gap. Therefore the live textbook keeps the Ward-identity mathematics but does not reuse the archived theorem as a current source of record.

This is a useful example of a general rule: a correct conditional field-theory identity can survive even when a larger phenomenological interpretation built on top of it is downgraded.

5.7 Vazba na theta kapitolu

The reduced theta kernel

$$S(t, \psi) = \sum_n a_n e^{-\pi\psi n^2} e^{i\pi t n^2}$$

provides a cleaner controlled laboratory for complex time than the old alpha renormalisation route. The same parameter ψ acts as a heat-time variable, while t carries the oscillatory phase. Fourier, Laplace, Mellin, and Poisson/Jacobi transforms can then be derived exactly before any physical UBT claim is attached to them.

Kapitola 6

Theta, Mellin, zeta, cesty a prvočíselná struktura

6.1 Proč tato kapitola existuje

This chapter is deliberately broader than a canonical UBT claim document. Its purpose is pedagogical: to place several classical mathematical structures next to the reduced UBT theta bridge so that their relations can be seen in one place. A result does *not* become a UBT theorem merely because it is useful inside UBT.

We therefore use three status labels in the prose:

- **Classical**: standard mathematics or standard quantum mechanics;
- **Bridge**: an exact statement about the reduced theta model used to connect standard mathematics to UBT notation;
- **UBT open**: a statement that would have to be derived from the canonical UBT field equations or action before it could become a physical UBT claim.

The central reduced object is

$$S(t,\psi) = \sum_{n\geq 1} a_n\, e^{-\pi\psi n^2} e^{i\pi t n^2} = \sum_{n\geq 1} a_n e^{-\pi(\psi - it)n^2}. \tag{6.1}$$

It is the same reduced complex-time kernel used in `canonical/bridges/theta_complex_time_bridge.tex`. In this chapter it is a **bridge model**; the canonical UBT master field $\Theta(q,\tau)$ is not assumed to be a Jacobi theta function.

Sign convention used in related RH notes. Some earlier UBT/RH scratch work used a positive heat variable $y = -\psi$ and therefore wrote the theta parameter as $t - i\psi = t + iy$ with convergence on the corresponding $y > 0$ half-line. The present chapter writes the damping variable directly as $\psi > 0$ in Eq. (6.1). The transform identities are identical after the relabelling $y = -\psi$; the sign convention must be declared before a physical UBT interpretation is attached to either half-line.

6.2 Jedno spektrum, několik reprezentací

6.2.1 Ortogonalita módů versus neortogonalita theta šablon

For fixed ψ, the elementary time modes in Eq. (6.1) are

$$e^{i\pi n^2 t}. \tag{6.2}$$

On the interval $t \in [0, 2]$ they are orthogonal:

$$\int_0^2 e^{i\pi(n^2-m^2)t}\,dt = 2\,\delta_{nm}, \qquad n, m \geq 1. \tag{6.3}$$

The important distinction is that the *complete templates* $S_\psi(t) := S(t, \psi)$ at different values of ψ are generally not orthogonal.

Věta 6.1 (Classical Gram kernel identity)**.** *For the inner product*

$$\langle f, g\rangle := \frac{1}{2}\int_0^2 f(t)\overline{g(t)}\,dt, \tag{6.4}$$

and amplitudes S_ψ of Eq. (6.1),

$$G(\psi, \phi) := \langle S_\psi, S_\phi\rangle = \sum_{n\geq 1} |a_n|^2 e^{-\pi(\psi+\phi)n^2}. \tag{6.5}$$

For $a_n \equiv 1$,

$$G(\psi, \phi) = \frac{\vartheta_3(0 \mid i(\psi + \phi)) - 1}{2}. \tag{6.6}$$

Důkaz. Insert the two sums in the inner product. Equation (6.3) eliminates all $m \neq n$ cross terms, leaving

$$\sum_n |a_n|^2 e^{-\pi\psi n^2} e^{-\pi\phi n^2}.$$

For unit weights, the standard identity $\vartheta_3(0 \mid iu) = 1 + 2\sum_{n\geq 1} e^{-\pi u n^2}$ gives Eq. (6.6). □

Practical consequence. Template non-orthogonality is not an obstacle to detection. If an observed signal is $f(t)$, a normalized matched-filter score is

$$\rho(\psi, \Delta) = \frac{\left|\int f(t)\overline{S(t-\Delta, \psi)}\,dt\right|^2}{\|f\|^2\,\|S_\psi\|^2}. \tag{6.7}$$

The best single template is obtained by maximizing ρ over (ψ, Δ). Orthogonality becomes important when many templates are fitted simultaneously; then the Gram matrix must be inverted or regularized, for example through $G + \lambda I$.

6.2.2 Fourierova transformace v reálném čase

Take the Fourier convention

$$\widehat{f}(\omega) = \int_{-\infty}^{\infty} f(t)e^{-i\omega t}\,dt. \tag{6.8}$$

Formally, or distributionally,

$$\widehat{S}(\omega, \psi) = 2\pi \sum_{n\geq 1} a_n e^{-\pi\psi n^2}\delta(\omega - \pi n^2). \tag{6.9}$$

Thus the quadratic mode spectrum is already a Dirac comb in real-time frequency; no mode-by-mode demodulation is needed to create the delta lines.

6.2.3 Laplaceova transformace v imaginárním čase

For $\operatorname{Re} z$ sufficiently large,

$$\mathcal{L}_\psi[S](z; t) = \int_0^\infty e^{-z\psi} S(t, \psi)\,d\psi \tag{6.10}$$

$$= \sum_{n\geq 1} \frac{a_n e^{i\pi t n^2}}{z + \pi n^2}. \tag{6.11}$$

The same eigenvalues that gave Fourier delta lines now appear as resolvent poles $z = -\pi n^2$.

6.2.4 Mellinova transformace v imaginárním čase

For $\mathrm{Re}\, s$ in the absolute-convergence half-plane,

$$\mathcal{M}_\psi[S](s;t) := \int_0^\infty \psi^{s-1} S(t,\psi)\, d\psi \tag{6.12}$$

$$= \Gamma(s)\pi^{-s} \sum_{n\geq 1} \frac{a_n e^{i\pi t n^2}}{n^{2s}}. \tag{6.13}$$

The identity follows from

$$\int_0^\infty \psi^{s-1} e^{-\pi n^2 \psi}\, d\psi = \Gamma(s)(\pi n^2)^{-s}. \tag{6.14}$$

For the Gram kernel with $a_n \equiv 1$ and $u = \psi + \phi$,

$$\int_0^\infty u^{s-1} G(u)\, du = \Gamma(s)\pi^{-s}\zeta(2s). \tag{6.15}$$

This is classical theta–Mellin mathematics. Its relevance here is organizational: the same reduced theta bank that causes non-orthogonal templates is also a heat trace whose Mellin representation is a spectral zeta function.

The spectral triangle. Equations (6.9), (6.11), and (6.13) give three views of the same quadratic spectrum:

$$\boxed{\begin{array}{lcl} \text{Fourier in } t & \longrightarrow & \delta(\omega - \pi n^2), \\ \text{Laplace in } \psi & \longrightarrow & (z + \pi n^2)^{-1}, \\ \text{Mellin in } \psi & \longrightarrow & \Gamma(s)\pi^{-s} n^{-2s}. \end{array}} \tag{6.16}$$

6.3 Mellin je Fourierova transformace v logaritmické souřadnici

Let

$$M_f(s) = \int_0^\infty f(x) x^{s-1}\, dx, \qquad s = \sigma + i\omega. \tag{6.17}$$

Set $x = e^u$, so $dx = e^u du$. Then

$$M_f(\sigma + i\omega) = \int_{-\infty}^\infty f(e^u) e^{\sigma u} e^{i\omega u}\, du. \tag{6.18}$$

Apart from the conventional sign of the Fourier phase, this is a Fourier transform in $u = \ln x$ of the weighted signal $f(e^u)e^{\sigma u}$. Thus Mellin analysis is the natural harmonic analysis for multiplicative scaling, whereas ordinary Fourier analysis is natural for additive translation.

This observation is especially useful when interpreting zeta functions: zeta's vertical variable $\mathrm{Im}\, s$ is a log-scale frequency.

6.4 Současný řez a časově fázovaná rodina zeta funkcí

UBT coordinate convention. In the working complex-time interpretation used in this research thread, the present slice is chosen as $t = 0$. This is a UBT coordinate convention, not a statement that the Riemann zeta function intrinsically knows a physical "present."

For unit weights $a_n \equiv 1$, define the quadratic time twist

$$Z_\Theta(s;t) := \sum_{n=1}^\infty \frac{e^{i\pi t n^2}}{n^{2s}}. \tag{6.19}$$

Then Eq. (6.13) becomes

$$\boxed{\mathcal{M}_\psi[S](s;t) = \Gamma(s)\pi^{-s} Z_\Theta(s;t).} \tag{6.20}$$

At the present slice,

$$\boxed{Z_\Theta(s;0) = \zeta(2s).} \tag{6.21}$$

Thus ordinary zeta is one member—the $t = 0$ member—of a natural family generated by the same reduced complex-time theta kernel.

Poznámka 6.2. The exponential factor in Eq. (6.19) is an *additive quadratic twist*. For generic t the resulting Dirichlet series does not have the ordinary Euler product. One must not infer prime-factor dynamics merely from the existence of the $t = 0$ Euler product.

6.4.1 Jednoduché speciální časy

The family is periodic under $t \mapsto t + 2$ because $n^2 \in \mathbb{Z}$:

$$Z_\Theta(s; t + 2) = Z_\Theta(s;t). \tag{6.22}$$

At $t = 1$,

$$Z_\Theta(s;1) = \sum_{n\geq 1} \frac{(-1)^{n^2}}{n^{2s}} = \sum_{n\geq 1} \frac{(-1)^n}{n^{2s}} \tag{6.23}$$

$$= -\eta(2s) = -(1 - 2^{1-2s})\zeta(2s), \tag{6.24}$$

where η is the Dirichlet eta function.

6.4.2 Racionální časy: rozklad pomocí Hurwitzovy zeta funkce

Let $t = a/q$ with integers a and $q > 0$. The coefficient

$$c_n = e^{i\pi a n^2/q} \tag{6.25}$$

is periodic with period $M = 2q$ (sometimes the true period is smaller). Splitting the Dirichlet series into residue classes gives the exact classical identity

$$\boxed{Z_\Theta\left(s; \frac{a}{q}\right) = M^{-2s} \sum_{r=1}^{M} e^{i\pi a r^2/q} \zeta\left(2s, \frac{r}{M}\right), \qquad M = 2q,} \tag{6.26}$$

where $\zeta(s, \alpha)$ is the Hurwitz zeta function.

Důkaz. Write every $n \geq 1$ uniquely as $n = mM + r$ with $m \geq 0$ and $1 \leq r \leq M$. Periodicity gives $c_{mM+r} = c_r$, and

$$Z_\Theta(s; a/q) = \sum_{r=1}^{M} c_r \sum_{m=0}^{\infty} (mM + r)^{-2s}$$

$$= M^{-2s} \sum_{r=1}^{M} c_r \zeta\left(2s, \frac{r}{M}\right).$$

□

This makes the “alternative zeta functions” generated away from $t = 0$ precise. The ordinary zeta, eta, Hurwitz zeta combinations, and quadratic Gauss sums are not unrelated objects here: they are different slices or representations of one quadratic-phase family.

6.5 Racionální revivaly a kvadratické Gaussovy součty

At rational $t = a/q$, the phase $e^{i\pi a n^2/q}$ depends only on a finite residue class. The corresponding finite quadratic sums are Gauss sums. A common normalization is

$$g(a,q) = \sum_{r=0}^{q-1} e^{2\pi i a r^2/q}. \tag{6.27}$$

For coprime moduli q_1, q_2, the Chinese remainder theorem gives, with the normalization in Eq. (6.27),

$$g(a, q_1 q_2) = g(a q_2, q_1)\, g(a q_1, q_2), \qquad \gcd(q_1, q_2) = 1. \tag{6.28}$$

Thus composite denominators factor into coprime pieces, and prime powers are the irreducible arithmetic building blocks of rational-time revival structure.

This is a more promising place to search for a *dynamical* origin of UBT prime sectors than simply observing the Euler product of $\zeta(2s)$, because the Gauss-sum arithmetic is carried directly by the quadratic theta phase.

6.6 Součet theta módů a Feynmanův součet windingů

The theta/path relation is exact in a controlled textbook example: a free particle of mass m moving on a circle of circumference L.

The momentum eigenfunctions and energies are

$$\phi_n(x) = L^{-1/2} e^{2\pi i n x/L}, \qquad E_n = \frac{\hbar^2}{2m}\left(\frac{2\pi n}{L}\right)^2. \tag{6.29}$$

Hence the real-time propagator is

$$K(x, x'; T) = \frac{1}{L} \sum_{n\in\mathbb{Z}} \exp\left(\frac{2\pi i n(x - x')}{L} - \frac{i E_n T}{\hbar}\right). \tag{6.30}$$

This is a Jacobi-theta mode sum. Poisson summation transforms it into

$$\boxed{K(x, x'; T) = \sqrt{\frac{m}{2\pi i \hbar T}} \sum_{w\in\mathbb{Z}} \exp\left[\frac{i m (x - x' + wL)^2}{2\hbar T}\right].} \tag{6.31}$$

The integer w labels winding sectors. The exponent is the classical free-particle action for the lifted path from x' to $x + wL$:

$$S_w^{\mathrm{cl}} = \frac{m(x - x' + wL)^2}{2T}. \tag{6.32}$$

Therefore the same propagator admits two exact representations:

$$\boxed{\text{theta/momentum modes} \quad \Longleftrightarrow \quad \text{Feynman winding histories}} \tag{6.33}$$

with Poisson/Jacobi transformation as the bridge.

Important qualification. The classical trajectory is selected by *stationary* action, $\delta S = 0$, not necessarily minimum action. Several stationary paths are ordinary quantum mechanics; by themselves they do not imply a multiverse interpretation.

6.7 Kde prvočísla vstupují a kde nevstupují

6.7.1 Eulerův součin na nefázovaném řezu

At $t = 0$ and unit weights,

$$Z_\Theta(s;0) = \zeta(2s) = \prod_p \left(1 - p^{-2s}\right)^{-1}, \qquad \operatorname{Re} s > \frac{1}{2}. \tag{6.34}$$

This is a profound change of representation from an additive sum over all positive integers to a multiplicative product over primes. It is nevertheless a classical property of the integers, not by itself evidence that UBT dynamics selects prime modes. For general weights a_n, the Dirichlet series $\sum |a_n|^2 n^{-2s}$ has an Euler product only when the coefficients satisfy the appropriate multiplicativity conditions.

6.7.2 Mřížky pólů lokálních Eulerových faktorů

A single formal local factor

$$L_p(s) = \left(1 - p^{-2s}\right)^{-1} \tag{6.35}$$

has poles when $p^{-2s} = 1$. Writing $s = \sigma + i\omega$ gives

$$\sigma = 0, \qquad \omega_{p,k} = \frac{\pi k}{\ln p}, \qquad k \in \mathbb{Z}. \tag{6.36}$$

For distinct primes $p \neq q$, nonzero pole frequencies cannot coincide. If $k/\ln p = \ell/\ln q$ with nonzero integers k, ℓ, then $q^k = p^\ell$, contradicting unique factorization. All local factors share the trivial $k = 0$ location.

These are poles of the *individual formal Euler factors*. They lie outside the Euler-product convergence half-plane and must not be confused with poles of the analytically continued global $\zeta(2s)$, whose pole is at $s = 1/2$.

If $N(T; P)$ counts positive local-factor pole frequencies up to height T for primes $p \leq P$, then

$$N(T;P) = \sum_{p \leq P} \left\lfloor \frac{T \ln p}{\pi} \right\rfloor \tag{6.37}$$

$$= \frac{T}{\pi}\vartheta(P) + O(\pi(P)), \tag{6.38}$$

where

$$\vartheta(P) = \sum_{p \leq P} \ln p \tag{6.39}$$

is Chebyshev's theta function. The prime number theorem implies $\vartheta(P) \sim P$.

6.7.3 Vertikální Fourierovo spektrum a mocniny prvočísel

At fixed $\sigma > 1/2$,

$$\zeta\big(2(\sigma + i\omega)\big) = \sum_{n \geq 1} n^{-2\sigma} e^{-i(2\ln n)\omega}. \tag{6.40}$$

Thus Fourier transformation in ω produces spectral lines at

$$\xi_n = 2 \ln n. \tag{6.41}$$

To isolate primes and prime powers, use the logarithmic derivative:

$$-\frac{\zeta'(z)}{\zeta(z)} = \sum_{n \geq 1} \frac{\Lambda(n)}{n^z} = \sum_p \sum_{k \geq 1} (\ln p) p^{-kz}, \qquad \operatorname{Re} z > 1. \tag{6.42}$$

With $z = 2(\sigma + i\omega)$,

$$-\frac{\zeta'(2(\sigma+i\omega))}{\zeta(2(\sigma+i\omega))} = \sum_p \sum_{k\geq 1} (\ln p) p^{-2k\sigma} e^{-i(2k\ln p)\omega}. \tag{6.43}$$

The prime-power frequency comb therefore sits at

$$\boxed{\xi_{p,k} = 2k \ln p.} \tag{6.44}$$

Distinct prime-power lines coincide only when the prime powers themselves are identical.

6.7.4 Počítání prvočísel a frekvence nul zeta funkce

The weighted prime-power counting function

$$\psi_C(x) = \sum_{n\leq x} \Lambda(n) \tag{6.45}$$

is related by the classical explicit formula to the nontrivial zeros $\rho = \beta + i\gamma$ of ζ. Schematically,

$$\psi_C(x) = x - \sum_\rho \frac{x^\rho}{\rho} + \text{elementary terms.} \tag{6.46}$$

Writing $x = e^u$ turns each zero contribution into

$$x^\rho = e^{\beta u} e^{i\gamma u}. \tag{6.47}$$

Thus the imaginary parts γ of zeta zeros act as oscillation frequencies in the logarithmic variable $u = \ln x$. This is the classical primes–zeros spectral bridge.

6.8 Co to znamená pro prvočíselné sektory UBT

Two distinct mechanisms must not be conflated.

Route A: trace/orbit route. Trace-formula analogies suggest that primes should correspond to primitive periodic objects whose repeated traversals generate prime powers. For the ordinary free particle on S^1, however, the primitive-orbit length spectrum is not $\{\ln p\}$. Therefore

> **GAP-THETA-PRIME-1:** derive from a genuine UBT operator a primitive-orbit or spectral structure whose natural primitive data are $\ln p$, or do not claim that the Euler product dynamically derives UBT prime sectors.

Route B: rational-revival route. The quadratic theta dynamics itself has arithmetic structure at rational times. Gauss sums factor through coprime denominators, making prime powers natural building blocks. This motivates

> **GAP-THETA-PRIME-2:** rewrite the existing UBT α prime-sector construction in revival/winding variables and test whether its distinguished primes arise from rational-time Gauss-sum structure rather than from a post-selected list.

These routes are compatible as research directions but are not yet known to be the same mechanism.

6.9 Operátorová otázka za celým obrazem

Equation (6.1) can be written formally by introducing

$$H_0|n\rangle = \pi n^2 |n\rangle. \tag{6.48}$$

Then its mode evolution is generated by

$$U_0(t,\psi) = e^{-\psi H_0} e^{+itH_0}. \tag{6.49}$$

This makes the reduced theta amplitude genuinely propagator-like. The decisive UBT question is stronger:

> **GAP-THETA-PROP:** derive an operator H_Θ from the canonical UBT action or field equations such that an appropriate kernel or matrix element of $e^{-\psi H_\Theta} e^{\pm itH_\Theta}$ yields the theta structure used here.

Until this gap is closed, the mode/winding, Mellin/zeta, and revival/prime relations are rigorous mathematics of the reduced bridge model, not a derivation of UBT dynamics.

6.10 Mapa velkého obrazu

The useful synthesis is therefore

$$\boxed{\begin{array}{ccc}
 & \text{reduced theta kernel} & \\
 & \sum_n a_n e^{-\pi\psi n^2} e^{i\pi t n^2} & \\
\swarrow & \downarrow & \searrow \\
\text{Fourier/Laplace} & \text{Mellin} & \text{Poisson/Jacobi} \\
\delta\text{-lines/resolvent} & Z_\Theta(s;t) & \text{winding/path sum} \\
 & \downarrow\ t=0 & \downarrow\ t\in\mathbb{Q} \\
 & \zeta(2s) & \text{Gauss/revival arithmetic} \\
 & \downarrow & \\
 & \text{Euler product / prime powers / zeros} &
\end{array}} \tag{6.50}$$

The classical arrows are exact. The research task is to determine which of them survive when the reduced kernel is replaced by a canonical UBT operator and whether either prime route explains the existing UBT α sectors.

Část III

Fenomenologie, verifikace a aplikace

Kapitola 7

Program konstanty jemné struktury: co je odvozeno a co není

7.1 Status před algebrou

The authoritative repository status is deliberately stronger than any elegant formula in this chapter:

> The physical fine-structure constant is not yet derived from first principles in UBT. The integer-137 route is structural/conditional, while the bridge fixing the required effective coefficient from the UBT action remains open.

The live sources of status are `STATUS.md`, `WHAT_IS_PROVED.md`, and `canonical/alpha/ALPHA_MASTER_STATUS.m` Historical texts using the phrase "fit-free alpha derivation" do not override these ledgers.

7.2 Podmíněný winding potenciál

A central reduced model studies

$$\boxed{V_{\text{eff}}(n) = n^2 - Bn\ln n, \qquad n > 0.}$$

Treating n temporarily as a continuous variable,

$$\frac{dV_{\text{eff}}}{dn} = 2n - B(\ln n + 1).$$

A stationary point therefore satisfies

$$\boxed{B = \frac{2n}{\ln n + 1}.}$$

If one asks for a stationary point at $n = 137$, the required coefficient is

$$B_{137} = \frac{274}{\ln 137 + 1} \approx 46.284.$$

This equation is an exact property of the reduced potential. It does *not* explain why the UBT action must generate that value of B.

The second derivative is

$$\frac{d^2V_{\text{eff}}}{dn^2} = 2 - \frac{B}{n}.$$

At a stationary point this becomes

$$2 - \frac{2}{\ln n + 1} = \frac{2\ln n}{\ln n + 1} > 0 \qquad (n > 1),$$

so the continuous stationary point is a local minimum. Integer and prime-sector restrictions can then be studied as an additional discrete selection problem.

7.3 Co může a nemůže znamenat "prvočíselná stabilita"

The repository contains a structural prime-stability analysis in which the integer sector around 137 is examined under a specified selection rule. A prime-labelled minimum can be a reproducible result of that reduced model. However:

- the primality of 137 is standard number theory;
- selection of a prime sector does not by itself identify α^{-1} with that sector;
- the physical correction from 137 to $137.036\ldots$ needs a separate derivation;
- the coefficient B must still be obtained from the UBT dynamics rather than inferred from the target value.

The theta/Mellin/Feynman chapter develops possible mechanisms for understanding where multiplicative prime structure could enter, but those mechanisms are research bridges, not completed alpha proofs.

7.4 Audit N_{eff}: dvě různá počítání se nesmějí zaměňovat

A previous derivation used

$$N_{\text{eff}} = 3 \times 2 \times 2 = 12.$$

The current audit distinguishes at least two objects:

$$\begin{aligned} N_{\text{eff}}^{\text{twist}} &= 12, && \text{in the SU(2)-twist construction,} \\ N_{\text{eff}}^{\text{loop}} &= 3, && \text{in the direct audited scalar-action loop count.} \end{aligned}$$

The identification

$$N_{\text{eff}}^{\text{loop}} = N_{\text{eff}}^{\text{twist}}$$

is not presently a proved theorem. Therefore a formula that inserts 12 into a loop coefficient must name the twist assumptions; it cannot be described as a unique consequence of $\mathbb{C} \otimes \mathbb{H}$ alone.

Correspondingly, the repository records two baseline coefficients in different routes,

$$B_0^{\text{loop}} = 2\pi, \qquad B_0^{\text{twist}} = 8\pi,$$

with the bridge from either baseline to the phenomenologically required $B \approx 46.284$ still open.

7.5 Gap G137-B

The decisive alpha problem can be stated without rhetoric:

> Starting from the canonical UBT action and its justified field content, derive the effective reduced coefficient multiplying $n \ln n$, including all normalisations and multiplicities, and show that the derivation is independent of the measured value of α.

This is Gap G137-B. Several historical and research-track routes have produced interesting modular, eta, Gamma, determinant, and winding structures, but the current master status records the required insertion into the physical coefficient as open.

7.6 Proč má stále smysl studovat theta/Mellinovu větev

The failure of the naive chain

$$\Theta \to \text{Mellin} \to \zeta \to \text{Euler product} \to \text{prime sectors}$$

to *dynamically* select primes is itself useful. The Euler product of a standard theta Mellin transform reflects multiplicative arithmetic already present in the integer index set. A stronger mechanism would have to identify primitive sectors dynamically.

Rational-time theta revivals are one candidate because quadratic Gauss sums factor over coprime denominators. In that setting prime powers are genuinely irreducible building blocks of the revival arithmetic. Whether the alpha prime-sector construction can be rewritten in those variables is an explicit open question, not an established result.

7.7 Bezpečné shrnutí

The statements to remember are:

1. the reduced stationary equation and its stability calculus are exact;
2. the integer/prime structural selection can be studied reproducibly;
3. B is not yet derived from the UBT action at the required value;
4. the relevant multiplicity bookkeeping is not fully reconciled;
5. $\alpha^{-1} = 137.036\ldots$ is not a first-principles UBT result today.

This is intentionally less dramatic than older archive text and more useful for future work, because it tells us exactly which equation must be closed next.

Kapitola 8

Testy, reprodukovatelnost a formální ověřování

Živou ověřovací vrstvou je aktivní repozitář, nikoli `ARCHIVE/`. Důležité vstupní body jsou:

- `tests/` pro vědecké, provenance a architektonické regresní testy;
- `tools/apply_provenance_headers.py` pro značení AI provenance;
- `tools/latex_audit.py` pro izolované překlady LaTeX dokumentů;
- `tools/check_alpha_claims.py` a aktivní alpha testy pro kontrolu tvrzení citlivých na aktuální status;
- `research_tracks/theta_spectral/theta_zeta_probe.py` pro numerické kontroly theta/Mellin/Gausso větve;
- `docs/DERIVATION_VERIFICATION_POLICY.md` jako závazná pravidla kontroly odvození v paperech.

8.1 Čtyři různé druhy kontroly

Je důležité nezaměňovat různé úrovně ověření. Překlad LaTeXu pouze dokazuje, že je dokument syntakticky sestavitelný. Numerický test ověřuje konkrétní číselné důsledky v dané přesnosti. CAS nástroj, například SymPy nebo Maxima, ověřuje symbolickou identitu, kterou jsme mu skutečně zadali. Formální systém Lean kontroluje, zda formalizovaný závěr plyne z formalizovaných předpokladů. Ani jedna z těchto kontrol sama o sobě nedokazuje, že příslušné předpoklady jsou fyzikálně vybrány kanonickou UBT dynamikou.

8.2 Lean jako preferovaný formální kontrolor

Pro nová nebo podstatně změněná teorémová odvození je preferovaným nástrojem Lean. Pokud je tvrzení rozumně formalizovatelné, cílem je mít vedle analytického odvození také kompilující `.lean` důkaz. Jestliže formalizace ještě není hotová nebo v daném prostředí Lean není dostupný, musí být stav výslovně označen jako `LEAN-PENDING`; nesmí se napsat, že tvrzení je "formálně dokázáno" jen proto, že agent vygeneroval Lean syntaxi.

Lean nenahrazuje ostatní kontroly. U hlavního výsledku paperu je žádoucí druhý, nezávislý kanál: například SymPy/Maxima pro symbolickou algebru a NumPy/Octave/MATLAB/Mathcad pro numerickou reprodukci. Dvě implementace by měly pokud možno používat odlišnou formulaci, aby pouze neopakovaly stejnou chybu.

8.3 Povinný záznam verifikace paperu

Každý nový nebo podstatně změněný paper má obsahovat sekci *Verification* nebo odkaz na doprovodný záznam. Pro hlavní tvrzení se zapisuje alespoň:

- identifikátor tvrzení, rovnice nebo lemmatu;
- analytický status a předpoklady;
- použitý nástroj a jeho verze;
- cesta k reprodukčnímu skriptu, testu, worksheetu nebo `.lean` souboru;
- výsledek a případná tolerance;
- přesně co kontrola ověřuje a co naopak neověřuje;
- stav Lean formalizace: `PROVED`, `PARTIAL`, `LEAN-PENDING` nebo `NOT-APPLICABLE`.

8.4 CI učebnice

Zaměřený workflow je `.github/workflows/textbook.yml`. Spouští se při relevantním pushi/pull requestu i ručně a používá `tools/latex_audit.py -strict`. Chyba kteréhokoli kořenového dokumentu učebnice tedy způsobí selhání workflow a neschová se v úspěšném non-strict auditu.

Po strict auditu se volá stejný Makefile jako lokálně a vzniká stabilní sada PDF v `build/textbook/public/`:

- `UBT_Textbook.pdf` — hlavní učebnice;
- `UBT_Textbook_Supplement_Covariant_Tetrad.pdf`;
- `UBT_Textbook_Supplement_Indexy_Torze_Rot_CS.pdf`.

GitHub Actions artefakt se jmenuje `ubt-textbook-pdfs-${run_id}` a obsahuje tato PDF i LaTeX audit.

Pro lokální publikovanou kopii lze spustit

```
make -C docs/textbook publish
```

což zkopíruje stejnou trojici do `docs/pdfs/`. Vygenerované PDF je pouze výstup; vědecký status zůstává řízen zdrojovým textem a provenance/status registry.

8.5 Minimální lokální kontrola před textbook patchem

Doporučená sekvence je:

```
python tools/apply_provenance_headers.py --apply
python tools/apply_provenance_headers.py --check
python tools/latex_audit.py --include 'docs/textbook/*.tex' \
  --include 'docs/textbook/**/*.tex' --jobs 2 --timeout 240 --strict
make -C docs/textbook verify
```

Úspěšný build potvrzuje syntaktické a reprodukční vlastnosti. Nezvyšuje sám od sebe Tier-C tvrzení na dokázaný výsledek UBT.

Kapitola 9

Aplikace a krátkodobé predikce

Část IV

Referenční dodatky

Příloha A

Formální důkazy a lemmata

Příloha B

FAQ pro recenzenty a inženýry

Příloha C

Spekulativní rozšíření (číst odděleně)